Borneo Hack 2025

ASEAN Trade Compass

AI-Powered Cross-Border Trade Guidance for ASEAN MSMEs

SDG 8: Decent Work & Economic Growth

SDG 9: Industry, Innovation & Infrastructure

SDG 10: Reduced Inequalities

Live deployment : https://borneohack-frontend.vercel.app/

Rookie

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March 2025

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1. Introduction

1.1 Case Study Background

Micro, Small, and Medium Enterprises (MSMEs) form the backbone of the ASEAN economy, accounting for more than 97% of businesses and contributing significantly to employment across the region. However, many MSMEs face challenges when attempting to expand beyond their local markets. Despite their large presence, their contribution to regional exports and GDP remains relatively low due to limited digital capabilities, lack of access to information, and difficulties navigating complex cross-border trade regulations.

Small vendors, particularly in regions such as Borneo, often struggle to understand export procedures, tariff structures, certification requirements, and international trade documentation. Language barriers and unfamiliar regulatory systems across ASEAN countries further complicate the process. As a result, many MSMEs are unable to participate fully in regional trade opportunities.

To address this issue, the project introduces **ASEAN Trade Compass**, an AI-powered platform that provides intelligent guidance on cross-border ASEAN trade regulations. Using technologies such as LangChain, retrieval-based AI, and translation tools, the platform allows MSMEs to ask trade-related questions, receive regulatory guidance, generate compliance checklists, and access essential documentation requirements to facilitate cross-border exports.

1.2 Chosen SGD

<u>SDG 8: Decent Work and Economic Growth</u>

SDG 8 focuses on promoting sustainable economic growth and productive employment opportunities. Many MSMEs in ASEAN struggle to expand their businesses due to limited knowledge of cross-border trade regulations, tariffs, and required documentation. The ASEAN Trade Compass platform helps MSMEs access clear trade information, compliance checklists, and AI-powered guidance, enabling them to participate more easily in regional trade. This supports business growth, job creation, and stronger economic participation.

<u>SDG 9: Industry, Innovation, and Infrastructure</u>

SDG 9 promotes innovation, technological advancement, and the development of reliable infrastructure. The proposed solution uses AI technologies such as LangChain, document retrieval systems, and translation tools to create a digital platform that simplifies complex trade regulations. By providing MSMEs with access to innovative digital tools, the platform encourages technological adoption and supports the development of smarter trade systems within ASEAN.

SDG 10: Reduced Inequalities

SDG 10 aims to reduce inequalities by ensuring equal access to opportunities. Many small businesses lack the resources and knowledge that larger corporations have when entering international markets. The ASEAN Trade Compass platform helps bridge this gap by providing MSMEs with accessible information and guidance for cross-border trade. This empowers smaller businesses to compete more fairly in regional markets and participate in the digital economy.

1.3 Problem Statement

Many MSMEs in ASEAN face difficulties in entering cross-border markets due to limited knowledge of export regulations, tariffs, certification requirements, and documentation procedures. These regulations differ across countries, making it difficult for small businesses to understand the steps required for compliance.

Additionally, language barriers and the lack of digital advisory tools prevent many vendors from effectively communicating with international buyers or understanding official trade documentation. Without accessible guidance and simplified information systems, MSMEs often remain restricted to local markets and miss opportunities for regional expansion.

Therefore, there is a need for an intelligent digital platform that can simplify trade regulations, provide real-time guidance, and assist MSMEs in navigating cross-border trade requirements within ASEAN.

1.4 Objectives

1. Provide AI-powered trade advisory support
   Develop an intelligent assistant that answers MSME queries related to export regulations, tariffs, and trade procedures.

2. Simplify regulatory compliance
   Generate clear step-by-step compliance checklists for exporting products between ASEAN countries.

3. Improve access to trade documentation
   Help users identify required documents such as Form D certificates, phytosanitary certificates, invoices, and shipping documents.

4. Reduce language barriers in cross-border trade
   Enable automatic translation and generation of trade communication to facilitate interaction with foreign distributors.

5. Promote digital adoption among MSMEs
   Encourage small businesses to utilize AI-powered tools to support decision-making and market expansion.

2. Overview of Existing Solutions

Several platforms and portals currently exist to support MSME trade and compliance. They can broadly be grouped into four categories:

2.1 Government & Intergovernmental Trade Portals

The ASEAN Business Advisory Council (ASEAN-BAC) Portal and the ITC SME Trade Help Desk (International Trade Centre) represent the primary intergovernmental resources available to regional businesses. The ASEAN-BAC portal provides policy information and trade statistics, but its content is entirely static and not tailored to the specific needs of individual businesses — offering no AI assistance, no document generation, and no real-time regulatory guidance. Similarly, the ITC SME Trade Help Desk offers trade manuals, export guides, and market access information targeted at developing-country SMEs globally, but lacks any conversational AI capability or personalized compliance tools, and is not tailored to the ASEAN context. At the national level, the Malaysia MITI and MATRADE Export Portal lists export procedures, incentives, and Form D guidance, but requires significant user expertise to navigate and provides no automation or AI-driven assistance.

2.2 Commercial SME Trade & E-Commerce Platforms

Alibaba.com Trade Assurance is the most prominent global SME marketplace, offering buyer-seller matching, payment protection, and logistics support, with ML-based product

recommendations and demand forecasting. However, it does not assist with export compliance, tariff classification, or certificate-of-origin documentation the very challenges facing Borneo MSMEs. Its high transaction fees and complex interface further make it unsuitable for first-time exporters. TradeGecko, rebranded as QuickBooks Commerce, addresses inventory management and order fulfillment for retailers but is limited to domestic commerce operations with no support for ASEAN cross-border compliance.

2.3 Domestic SME Digitization Tools

TradeEasy, a Malaysian platform, assists SMEs with export documentation and HS code lookup, but does not cover multi-country ASEAN trade routes or address the specific challenges faced by Borneo-based exporters. It lacks AI-driven regulation Q&A, Form D eligibility checking, and multilingual support. MyExport, an initiative by the Malaysia Digital Economy Corporation (MDEC), focuses on digitization support and grants to help SMEs establish an online presence, but is primarily an e-commerce onboarding programme rather than a cross-border trade compliance solution.

2.4 Competitive Landscape

The table below summarizes the key characteristics and limitations of existing solutions compared to our proposed BorneoTrade AI:

Solution	Target Users	Key Features	AI/ML Used	Languages	Gaps & Limitations
TradeEasy (Malaysia)	Malaysian SMEs	Export docs, HS code lookup	Basic rule-based	EN, BM	No ASEAN cross-border focus; no conversational AI; no Form D checker
Alibaba.com Trade Assurance	Global SMEs	Payments, buyer-seller matching, logistics	ML recommendations	Multi language	No compliance guidance; complex UI; high fees; unsuitable for micro-enterprises
ITC SME Trade Help Desk	Developing-country SMEs	Trade info portal, export guides, manuals	None	EN, FR, ES	Entirely static; no dynamic AI; not tailored to ASEAN or Borneo context

ASEAN-BAC Trade Portal	ASEAN businesses	Policy info, trade statistics	None	EN only	No real-time guidance; no document generation; no personalization
TradeGecko / QuickBooks Commerce	Retailers & Wholesalers	Inventory, orders, invoicing	Basic forecasting	EN only	No export compliance support; no ASEAN regulatory navigation
Asean Trade Compass (Our Solution)	Borneo MSMEs	AI Q&A, compliance checklist,The ASEAN Form D check, letter generation	LLM + RAG + Vector DB	EN, BM	Currently limited to ingested PDFs; expanding coverage is straightforward

2.5 Identified Gaps and Limitations

Based on the review above, the following critical gaps exist in the current landscape:

2.5.1 No ASEAN-Specific, AI-Driven Compliance Guidance

Existing tools either cover global trade generically or focus narrowly on a single country. None provide an intelligent, conversational interface that guides Borneo-based MSMEs through the specifics of ASEAN cross-border trade, including ATIGA preferential tariff rates, Form D certificate-of-origin requirements, and destination-country import regulations for specific products.

2.5.2 No Localized, Multilingual Support for Borneo Vendors

Borneo's business community includes vendors who speak Bahasa Malaysia, various indigenous languages, and limited English. Existing platforms are predominantly English-only. This language barrier prevents informal and micro-enterprises from independently navigating trade regulations, effectively excluding them from the benefits of regional economic integration.

2.5.3 Absence of Automated Document Generation

No existing solution enables small vendors to automatically generate trade compliance documents such as export inquiry letters, product-specific compliance checklists, or Form D eligibility assessments. Vendors currently rely on expensive trade consultants or must manually interpret government circulars — a process that is error-prone and time-consuming for micro-enterprises.

2.5.4 Static Knowledge Bases Without RAG or LLM Integration

Government portals and trade help desks rely on static PDFs, FAQs, and manuals. These resources are not searchable by natural language, do not update dynamically, and cannot answer context-specific questions. The application of Retrieval-Augmented Generation (RAG) combined with a large language model as implemented in our solution is absent from all identified existing tools in the ASEAN MSME space.

2.5.5 No Integration of Form D Eligibility Assessment

The ASEAN Trade in Goods Agreement (ATIGA) Form D certificate is critical for MSMEs to qualify for 0% preferential tariff rates across ASEAN. However, no existing digital tool provides an automated eligibility checker that walks vendors through the Rules of Origin criteria and assesses whether their product qualifies. This gap results in MSMEs either missing out on tariff benefits or submitting incorrect documentation.

2.5.6 High Cost and Complexity Barriers

Enterprise-grade platforms such as supply chain optimization tools and global marketplace platforms are priced for large corporations. The cost structure, onboarding complexity, and technical requirements create an insurmountable barrier for micro and small enterprises in rural Asean who may operate with minimal digital literacy.

2.6. Positioning of Our Solution

AI Trade Compass directly addresses the six gaps identified above through the following design decisions:

2.6.1 LLM + RAG Architecture

At the core of AI Trade Compass is a Retrieval-Augmented Generation (RAG) pipeline combining a large language model with a vector store ingested from official ASEAN trade documents and regulatory PDFs. Unlike static portals, the system dynamically retrieves the most relevant regulatory content and synthesizes it into a structured, plain-language answer

tailored to the user's specific question ensuring responses are grounded in verified official sources.

2.6.2 Product-Specific Compliance Checklists

AI Trade Compass generates compliance checklists tailored to both the product and destination country. The system first identifies the product category agricultural, electronics, textile, or otherwise and applies only the relevant regulations. For example, exporting durian to Thailand triggers phytosanitary and food safety requirements, while electronics require electrical safety certification. Each checklist item names the responsible agency and the required document a level of specificity absent from all reviewed competitors.

2.6.3 Form D Eligibility Assessment

The ATIGA Form D certificate of origin allows goods to qualify for preferential tariff rates of up to 0% under the ASEAN Free Trade Area. No existing competitor offers an automated eligibility tool for this. AI Trade Compass guides vendors through Rules of Origin criteria regional value content, product transformation, and HS code classification and delivers a clear eligibility determination with actionable guidance, saving vendors consulting fees and preventing costly customs errors.

2.6.4 Formal Letter Generation

Many Borneo MSMEs lack the business writing skills needed to communicate with government trade offices or overseas partners. AI Trade Compass automatically generates professional inquiry letters based on the vendor's product, destination country, and company details covering sender introduction, product description, import requirement requests, and compliance declarations removing the need for expensive trade consultants for basic business correspondence.

2.6.5 Multilingual Design

Borneo's business community is linguistically diverse, with many vendors more comfortable in Bahasa Malaysia than English. AI Trade Compass incorporates a dedicated translation chain that generates output checklists and letters in both languages. This is a deliberate architectural feature ensuring translated content is contextually accurate, making the platform genuinely accessible to the local MSME community rather than just digitally sophisticated users.

2.6.6 Accessibility-First Approach

AI Trade Compass is built to be lightweight and affordable. Its FastAPI backend and streamlined web frontend require no software installation and run on minimal server infrastructure, making it viable for NGOs, government agencies, or cooperatives to deploy at low cost in sharp contrast to the enterprise-grade platforms reviewed earlier, which demand dedicated IT teams and expensive licensing well beyond the means of small Borneo vendors.

3. Methodology

3.1 Technology Stack

Layer	Technology	Purpose
AI / LLM	Groq API (LLaMA 3.1 8B Instant)	Ultra-fast LLM inference for all AI features
RAG Framework	LangChain 0.3	RAG pipeline, chains, retrieval orchestration
Vector Database	Pinecone (Starter tier)	Store & search document embeddings at scale
Embeddings	all-MiniLM-L6-v2 (HuggingFace)	384-dim sentence embeddings for semantic search
Backend API	FastAPI 0.115 + Uvicorn	RESTful endpoints, async request handling
Frontend	React + Vite (TypeScript)	Mobile-first SPA with fast hot-reload dev build
Translation	Deep Translator (Google Translate API)	Multilingual letter output for ASEAN destinations
PDF Ingestion	PyPDFLoader + RecursiveCharacterTextSplitter	Chunk and embed trade regulation PDFs
Deployment	Railway (Procfile)	Cloud-native deployment via Procfile configuration

3.2 System Architecture

High-Level Architecture

The system follows a Retrieval-Augmented Generation (RAG) architecture. Trade regulation documents (PDFs) are ingested offline into a Pinecone vector store. At query time, the backend retrieves the most semantically relevant document chunks and injects them as grounding context into LLM prompts, ensuring factually anchored responses.

SYSTEM ARCHITECTURE OVERVIEW

React + Vite Frontend (Port 5173)

↕ HTTP / REST API

FastAPI Backend (Port 8000)

↕ LangChain RAG Chain

Pinecone Vector DB ↔ Groq LLM API ↔ HuggingFace Embeddings

↑ Offline Ingestion Pipeline

ASEAN Trade Regulation PDFs (data/ folder)

3.3 Data Flow

Ingestion (offline, one-time):

PDF files placed in the data/ directory are loaded using PyPDFLoader, which reads all pages from each document. The content is then split into overlapping chunks using RecursiveCharacterTextSplitter, with configurable chunk size and overlap to preserve context across splits. Each chunk is tagged with country metadata derived from the PDF filename for example, a file containing "malaysia" in its name is automatically labelled with metadata.country = "Malaysia". Embeddings are then generated for each chunk using the all-MiniLM-L6-v2 model, producing 384-dimension vectors that capture the semantic meaning of the text. Finally, these vectors are upserted into the Pinecone index in batches of 100 for efficient and reliable storage.

Query time (live):

When a user submits a query through the React frontend, it is sent as an HTTP POST request to the FastAPI backend. The query is then embedded using the same all-MiniLM-L6-v2 model used during ingestion, and a cosine similarity search is performed against the Pinecone index to retrieve the most relevant chunks (top-k). The retrieved chunks are formatted with their source and country metadata, then injected into a structured prompt template to provide the LLM with grounded context. The Groq API, powered by LLaMA 3.1 8B Instant, processes the prompt and generates a response anchored to the retrieved documents. Finally, both the AI-generated answer and the source citations are returned to the frontend for display to the user.

3.4 Implementation Details

API Endpoints

Endpoint	Method	Description
POST /api/query	POST	RAG-powered trade regulation Q&A
POST /api/checklist	POST	Generate export compliance checklist for a product + destination
POST /api/form-d-check	POST	Evaluate Form D / Certificate of Origin eligibility
GET /api/form-d-questions	GET	Return structured Form D interview questions (6 sections, 20 questions)
POST /api/translate-letter	POST	Generate + translate formal export business letter
POST /api/ingest	POST	Upload and ingest a new PDF into the vector store
GET /health	GET	Health check — returns version and status

3.5 Feature Implementation

Feature A — Trade Regulation Q&A

The QA chain uses a two-stage approach: first, a semantic retrieval step fetches the top-8 most relevant document chunks from Pinecone using cosine similarity. These are formatted with source/country metadata and injected into a structured prompt alongside a strict system message instructing the LLM to always provide concrete, actionable answers including HS codes, tariff rates, and agency names. Temperature is set to 0.3 to balance factual accuracy with natural language fluency.

Feature B — Export Compliance Checklist

The checklist chain includes an intelligent product category detector that classifies inputs as 'electronics', 'food_agricultural', or 'general' based on keyword matching. This allows the retrieval query to be specialised e.g., electronics queries target technical standards and certifications while food queries focus on phytosanitary and agricultural import rules. Electronics results are additionally filtered to exclude food-specific regulatory chunks.

Feature C — Form D Eligibility Assessment

The Form D chain conducts a 20-question structured interview across 6 sections (business info, product origin, raw materials, manufacturing process, regional value content, supporting documents). After collecting answers, a deterministic pre-screening calculates the Regional Value Content (RVC) using the formula: RVC = (FOB Price − Non-ASEAN Cost) / FOB Price ×

100%. The system assigns a preliminary verdict (ELIGIBLE / NOT ELIGIBLE / MORE INFORMATION REQUIRED) which is then passed to the LLM for a plain-language explanation with specific next steps and application guidance.

Feature D — Multilingual Letter Generator

The translation chain first generates a professional English export business letter via LLM, then uses the Deep Translator library (Google Translate backend) to produce a localised version in the destination country language. Long letters exceeding 4,500 characters are automatically chunked and translated segment by segment to respect API limits, then reassembled. The language mapping covers all 10 ASEAN member states.

4. Results

4.1 Summary of Outcomes

The platform delivered four fully working AI features by end of the hackathon:

• Trade Regulation Q&A — Vendors can ask any ASEAN export/import question and receive grounded answers sourced from real trade regulation documents.

• Export Compliance Checklist — The system generates a step-by-step checklist tailored to the product and destination country, with smart category detection for food and electronics.

• Form D Eligibility Checker — A 20-question guided interview evaluates whether a vendor qualifies for the ATIGA Certificate of Origin, including automatic Regional Value Content (RVC) calculation.

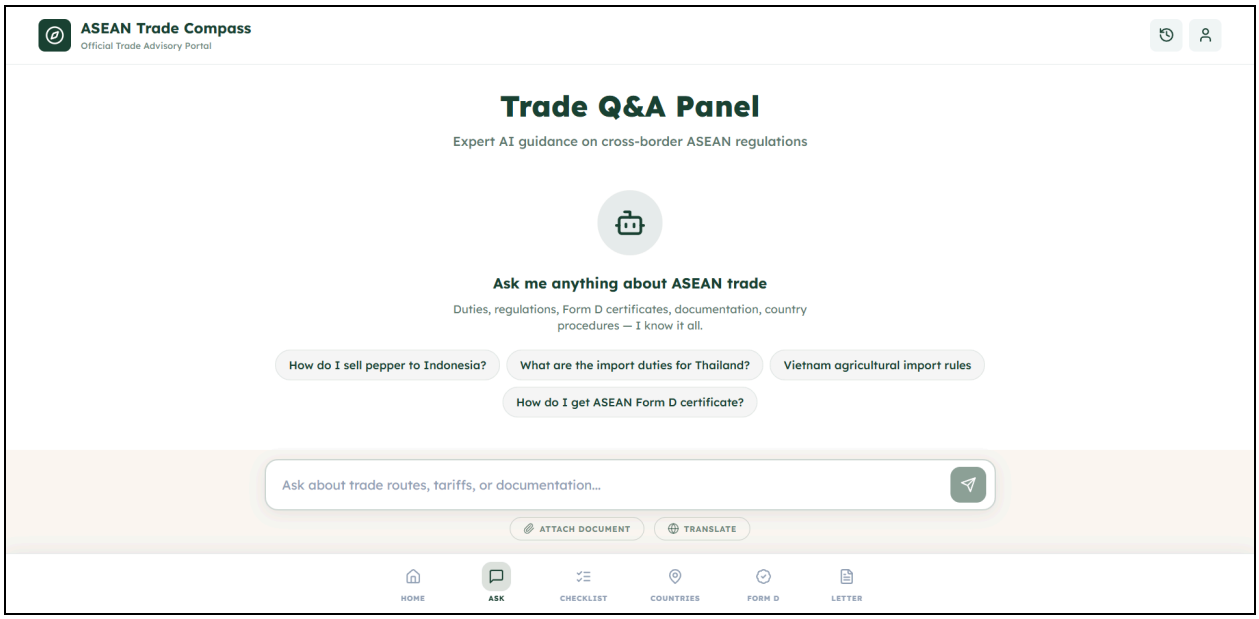
• Multilingual Letter Generator — Formal export business letters are generated in English and automatically translated into the destination country's language, covering all 10 ASEAN nations.

No.	Objective	Result	Status
1	Provide AI-powered trade advisory support — develop an intelligent assistant that answers MSME queries related to export regulations, tariffs, and trade procedures.	Implemented a RAG-based Q&A assistant using LangChain + Pinecone + Groq (LLaMA 3.1). Users can ask trade-related questions and receive grounded, document-backed answers with source citations.	✅ Met
2	Simplify regulatory compliance — generate clear step-by-step compliance checklists for exporting products between ASEAN countries.	Built a checklist generator that produces tailored, step-by-step export compliance checklists based on product type and destination country, with smart category detection for electronics and food products.	✅ Met
3	Improve access to trade documentation — help users identify required documents such as Form D certificates,	Developed a guided Form D eligibility checker with a 20-question interview. It calculates Regional Value Content (RVC), determines eligibility, and lists	✅ Met

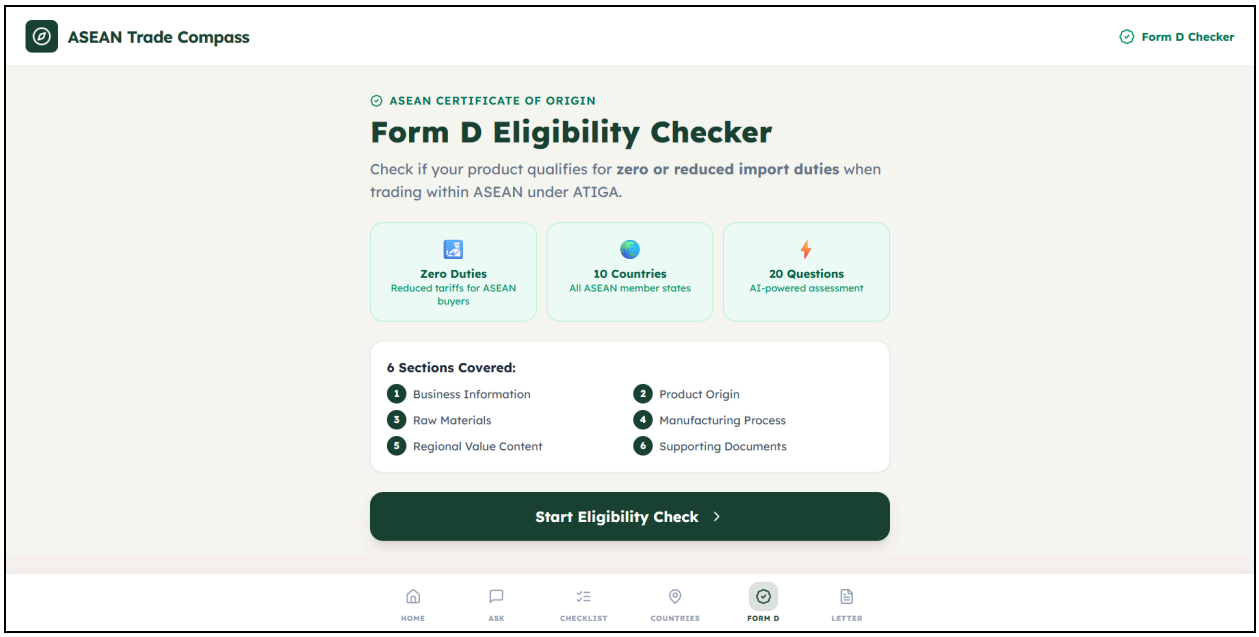
No.	Objective	Result	Status
	phytosanitary certificates, invoices, and shipping documents.	all required documents with application guidance.	
4	Reduce language barriers in cross-border trade — enable automatic translation and generation of trade communication to facilitate interaction with foreign distributors.	Created a multilingual letter generator that produces formal export business letters in English and automatically translates them into the destination country's language, covering all 10 ASEAN languages.	✅ Met
5	Promote digital adoption among MSMEs — encourage small businesses to utilize AI-powered tools to support decision-making and market expansion.	Delivered a mobile-first React application with a simple guided interface requiring no technical knowledge. Built on free-tier APIs (Groq, Pinecone) to keep the platform accessible and affordable for MSMEs.	✅ Met

5. Prototype Showcase

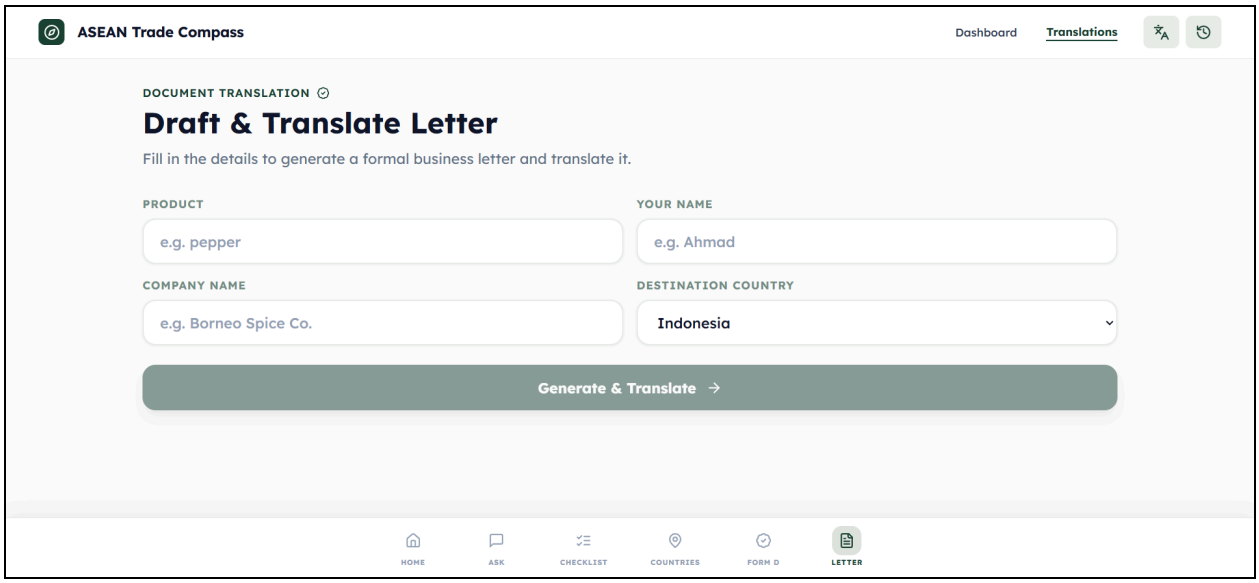
The central hub of the application, featuring an export-focused search bar, "Quick Trade Tips," and immediate access to trade actions. It also highlights "Trade Updates" (like the Borneo-Nusantara route) and provides a quick reference table for "Standard Documents" and "CO Issuing Authorities" across ASEAN.



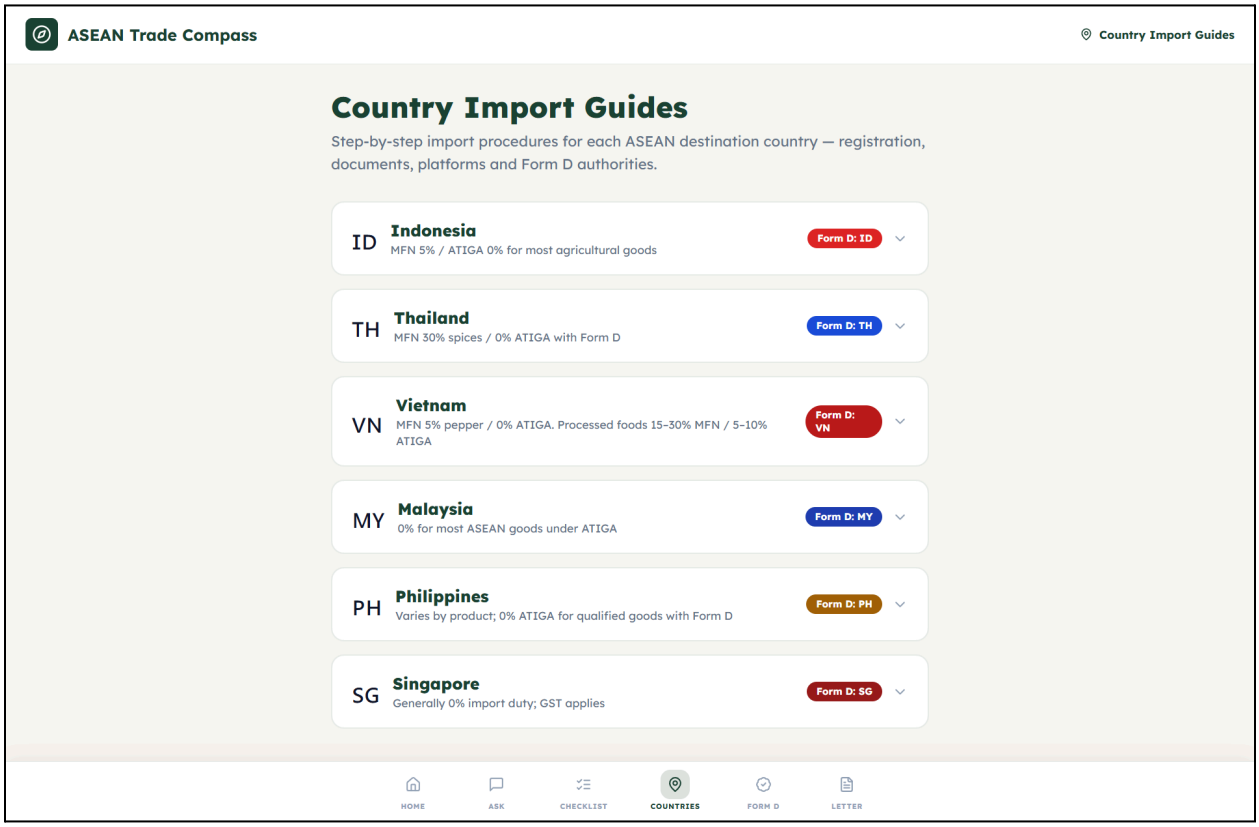
An AI-powered advisory portal designed to answer complex regulatory questions. It features a conversational interface where users can ask about duties, documentation, or specific regional rules, with additional options to attach documents or translate queries.



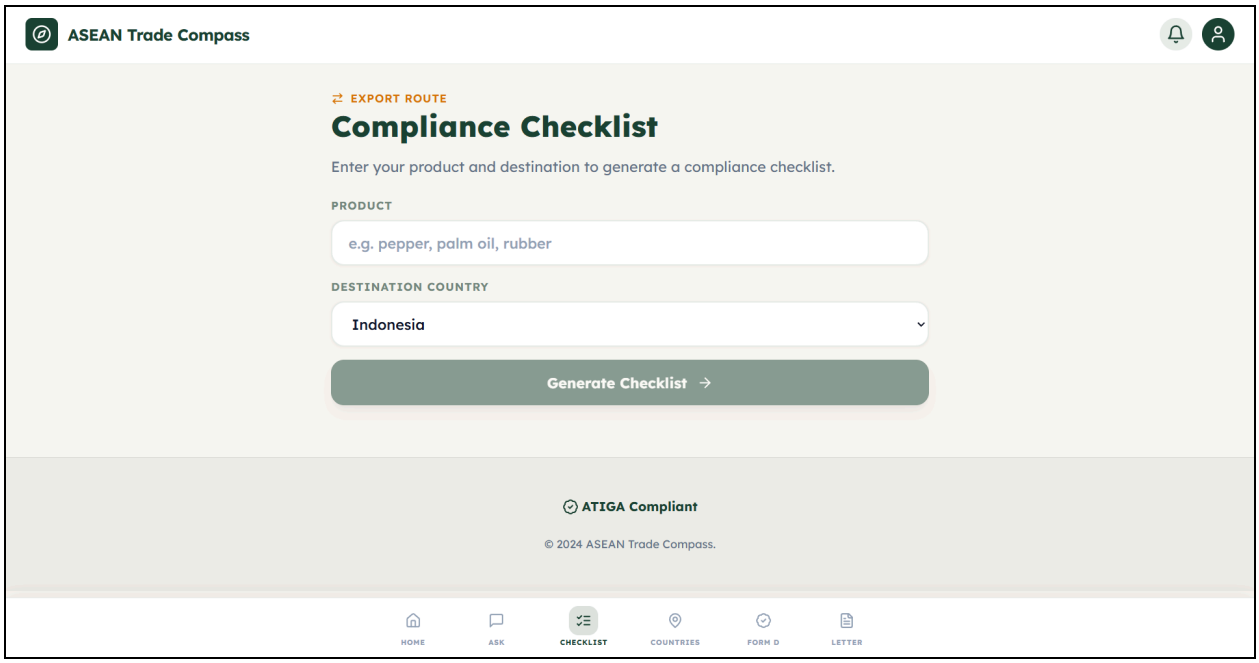
A structured, step-by-step assessment tool that determines if a product qualifies for zero or reduced import duties under the ASEAN Trade in Goods Agreement (ATIGA). It guides users through six critical sections, including product origin and regional value content.



A specialized tool for generating professional business correspondence. It allows users to input product and company details to automatically draft and translate formal letters into the language of the destination country, facilitating smoother cross-border communication.



A comprehensive directory providing a breakdown of import procedures for each ASEAN member state. It displays specific tariff information (such as MFN vs. ATIGA rates) and provides direct links to the relevant "Form D" authorities for each nation.



An interactive form where exporters can input their specific product and destination country to generate a tailored compliance roadmap. This tool ensures that all regulatory requirements are met before shipping, specifically tracking ATIGA compliance.

6. AI Acknowledgement

This section discloses all Artificial Intelligence tools used throughout the ideation, development, and finalization phases of the ASEAN Trade Compass project, in accordance with hackathon requirements.

6.1 Mandatory Disclosure Statement

The development team explicitly utilized AI tools as collaborative assistants during this project. AI was not used to fabricate data, generate false claims, or replace original human thinking. All AI-generated content was reviewed, verified, and guided by the team. The core idea, problem framing, system architecture decisions, and final integration were directed by human developers.

6.2 AI Tools Used

AI Tool	Category	How It Was Used	Scope
Claude (Opus 4.6) via Antigravity	Code Assistant	Core app development, RAG pipeline architecture, LangChain configuration,	Backend development &

		FastAPI backend logic, debugging and code review	system integration
Google Stitch	UI Generation	Generated the entire React + Vite frontend including component layout, styling, and mobile-responsive design	Frontend / React UI
Groq API (LLaMA 3.1)	LLM Inference	Powers the live AI assistant — processes user trade queries and generates responses grounded in retrieved document context	Core in-app AI feature
all-MiniLM-L6-v2 (HuggingFace)	Embedding Model	Converts trade regulation PDF text into vectors during ingestion, and converts user queries into vectors at runtime for semantic search	PDF ingestion & retrieval pipeline

6.3 Detailed Usage by Phase

Ideation & Architecture:

Claude was used to evaluate RAG architecture options, compare vector database choices, and validate the technical feasibility of combining LangChain with the Groq API. The team provided the problem context; Claude helped translate requirements into a concrete system design.

Backend Development:

Claude assisted with writing and debugging the FastAPI endpoints, LangChain RAG chain, Pinecone retriever setup, and prompt template engineering. All code was reviewed and tested by the team before integration.

Frontend Development:

Google Stitch generated the complete React + Vite frontend interface including the chat UI and mobile-responsive layout. The team specified the design requirements and user flow.

Document Ingestion & Embedding:

The all-MiniLM-L6-v2 model converted official ASEAN trade PDFs into 384-dimensional vectors stored in Pinecone for semantic retrieval. This enables grounded, document-backed answers rather than hallucinated responses.

Documentation & README:

Claude was used to draft project README.md, including setup instructions.The teams need to verity the API key guides, PDF ingestion steps, and troubleshooting tables. All content was verified against the actual project files before finalizing.

6.4 What AI Was NOT Used For

- AI was not used to generate or fabricate project results, outcomes, or research findings such as research data.
- AI was not used to write the problem statement, SDG alignment, or competitive analysis these reflect original research by the team.
- AI was not used to replace human decision-making on system design, feature prioritization, or project direction.

6.5 Responsible AI Usage Statement

All AI outputs were treated as drafts subject to human review. The team takes full responsibility for the correctness, completeness, and integrity of all content submitted as part of this project.

7. Conclusion

The ASEAN Trade Compass project demonstrates how artificial intelligence can be applied to simplify complex cross-border trade processes for Micro, Small, and Medium Enterprises (MSMEs) within the ASEAN region. Many small businesses struggle to expand internationally due to limited knowledge of export regulations, complicated documentation procedures, and language barriers. By integrating modern AI technologies such as Retrieval-Augmented Generation (RAG), vector databases, and large language models, the platform provides an intelligent and accessible solution that guides users through trade regulations and compliance requirements.

The system successfully delivers four core functionalities: AI-powered trade regulation question answering, export compliance checklist generation, Form D eligibility assessment, and multilingual export letter generation. Together, these features provide MSMEs with practical tools to understand export requirements, prepare necessary documentation, and communicate effectively with foreign partners. The use of document-grounded responses ensures that information provided to users is based on verified regulatory sources rather than generic AI outputs, improving reliability and trust.

The project also contributes to broader developmental goals by supporting digital adoption among small businesses and promoting fairer participation in regional trade. By lowering informational and technical barriers, the platform empowers smaller vendors especially those in regions such as Borneo to access opportunities that were previously difficult to navigate without specialized knowledge or expensive consulting services.

Despite the successful prototype developed during the hackathon, several areas for improvement remain. Future work could expand the knowledge base by ingesting more official trade regulation documents from all ASEAN member states to improve coverage and accuracy.

Integration with real-time government APIs could also allow the platform to provide updated tariff rates, customs requirements, and policy changes. Additionally, further development of multilingual capabilities and support for local dialects could enhance accessibility for a broader range of users. Other potential improvements include implementing automated HS code classification using machine learning models, integrating logistics and shipping service recommendations, and enhancing the user interface with mobile-first features designed specifically for micro-enterprises.

Overall, the ASEAN Trade Compass prototype demonstrates the strong potential of AI-powered advisory systems to transform how MSMEs interact with international trade regulations. With further development and expanded data coverage, the platform could evolve into a comprehensive digital trade assistant that helps small businesses confidently participate in ASEAN's growing regional economy.